COMP 2310: Discrete Structures
Midterm Review Guide

Question types

- **Multiple choice:** 8 questions, 24 points
- **Short computation and written questions:** 15 questions, 76 points, in three groups by chapter.
- Chapter 1 questions: set operations, counting with sets, gcd and lcm, matrix operation (23 points).
- Chapter 2 questions: truth table, quantifiers, converse and contrapositive, induction proof, exact cover (30 points).
- Chapter 3 questions: permutations, distinguishable arrangements, combinations, pigeonhole, probability, recurrence relations (23 points).

CHAPTER 1: FUNDAMENTALS

1.1 Sets

- Need to understand what a set is: distinct objects where order and repeats do not matter. Know the symbols $\in$, $\notin$, $\subseteq$, $\emptyset$ and $|A|$, and roster form $\{2, 4, 6, 8\}$ versus set-builder form $\{x \mid x \text{ is an even positive integer and } x < 10\}$.
- Need to be able to tell $\in$ from $\subseteq$. For $A = \{1, \{2, 3\}, 4\}$: $\{2, 3\} \in A$ is true, but $\{2, 3\} \subseteq A$ is false.

1.2 Set operations

- Need to know the operations: $A \cup B$ (in A or B), $A \cap B$ (in both), $A - B$ (in A but not B), A' (in U but not A), $A \oplus B$ (in exactly one). Write each set in roster form and work element by element.
- Need to know De Morgan: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

1.3 Addition principle (inclusion–exclusion)

- Need to know $|A \cup B| = |A| + |B| - |A \cap B|$, and that "neither" = total $-$ $|A \cup B|$. (Three sets: add the singles, subtract the pairs, add the triple.)
- Example: of 50 students, 28 like tea, 22 like coffee and 10 like both. At least one: $28 + 22 - 10 = 40$. Neither: 10.

1.4 Sequences and strings

- Need to understand explicit formulas ($a_n = 3n + 1$) versus recursive formulas, and regular expressions: * is zero or more repeats, $\vee$ is "or". For example $a(b \vee c) = \{ab, ac\}$.

1.5 gcd and lcm

- Need to be able to use Euclid's algorithm: replace (m, n) by (n, m mod n) until n = 0; the last m is the gcd. Then lcm = m × n / gcd. Example: gcd(84, 36) = 12 (84 mod 36 = 12, 36 mod 12 = 0), lcm = 252.

1.6 Matrices

- Need to know the size rules: add and subtract need the same size; AB needs (columns of A) = (rows of B) and is m × p for m × n times n × p; AB and BA are usually different; the transpose swaps rows and columns; an inverse needs AB = BA = I.
- Need to be able to multiply: entry (i, j) = row i of A times column j of B. Example: [1 0 ; 2 3] times [4 1 ; 0 2] = [4 1 ; 8 8]. For Boolean matrices use AND for multiply and OR for add.

1.7 Mathematical structures

- Need to understand closure, commutative, associative and distributive. Union and intersection are commutative; set difference, matrix multiplication and subtraction are not.

CHAPTER 2: LOGIC

2.1 Propositions and connectives

- Need to understand what a proposition is: a declarative statement that is true or false, not both. Questions, commands and "x + 4 = 9" are not propositions.

p	q	~p	p ∧ q	p ∨ q	p ⇒ q	p ⇔ q
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

- Need to know: AND is true only if both are true; OR is false only if both are false; p ⇒ q is false only when p is true and q is false; p ⇔ q is true when p and q match.

2.2 Truth tables

- Need to be able to build a truth table: 2^n rows for n variables; one column for each sub-expression, working from the inside of the brackets outward; the last column is the answer. Example: (p ∨ q) ∧ ~q gives F, T, F, F for rows TT, TF, FT, FF.
- Need to be able to find a truth value from given values. With p = T, q = F, r = T: (p ∧ q) ∨ r = F ∨ T = T.

2.3 Conditional statements

- Need to know the converse (q ⇒ p), the contrapositive (~q ⇒ ~p) and the inverse (~p ⇒ ~q) of p ⇒ q. Only the contrapositive is equivalent to the original; also p ⇒ q ≡ ~p ∨ q.
- **Important:** the contrapositive swaps AND negates. "If it rains, the ground is wet" has contrapositive "If the ground is not wet, it does not rain."

2.4 Tautology and equivalence

- Need to know that a tautology is true in every row (p ∨ ~p), a contradiction is false in every row (p ∧ ~p), and p ≡ q means both have the same final column. Test with a full truth table. Example: p ⇒ q and ~p ∨ q both give T, F, T, T.

2.5 Quantifiers

- Need to understand ∀x P(x) (every x) and ∃x P(x) (some x). To show ∀ is false, give ONE counterexample; to show ∃ is true, give ONE example. Negation: ~∀x P(x) ≡ ∃x ~P(x) and ~∃x P(x) ≡ ∀x ~P(x).
- Example: integers, Q(x): x + 2 < 5. ∀x Q(x) is false (x = 3); ∃x Q(x) is true (x = 0). Negation of "Every student passed": "Some student did not pass."

2.6 Mathematical induction

- Need to be able to write an induction proof: state P(n); basis step: check the first value of n; induction step: assume P(k) and, starting from the left side of P(k + 1), use the assumption to reach the right side; then conclude P(n) for all n.
- Example: 2 + 4 + ⋯ + 2n = n(n + 1). Basis: 2 = 1·2. Step: k(k + 1) + 2(k + 1) = (k + 1)(k + 2).

2.7 Exact cover

- Need to understand what an exact cover is: a collection of the subsets that are pairwise disjoint and whose union is A, so every element is in exactly one chosen subset.
- Need to be able to find one by choosing a subset for the element with the fewest options, removing the subsets that overlap it, and repeating (backtrack if stuck); then check both conditions.

CHAPTER 3: COUNTING

3.1 Multiplication principle and permutations

- Need to know: multiply the number of choices at each step. Order matters: n^r with repeats, $P(n, r) = n! / (n - r)!$ without repeats. Example: an ID of 3 letters then 2 digits, repeats allowed: $26^3 \times 10^2 = 1{,}757{,}600$.
- Need to know distinguishable arrangements: $n! / (k_1!\, k_2! \cdots)$. Example: SUCCESS has 7 letters with S three times and C twice, so $7! / (3!\, 2!) = 420$.

3.2 Combinations

- $C(n, r) = n! / (r!\,(n - r)!)$ when order does not matter, and $C(n + r - 1, r)$ when repeats are also allowed. Ask yourself two questions: does order matter, and may items repeat?

Order matters?	Repeats allowed?	Formula
Yes	Yes	n^r
Yes	No	$P(n, r) = n! / (n - r)!$
No	No	$C(n, r) = n! / (r!\,(n - r)!)$
No	Yes	$C(n + r - 1, r)$

- Examples: a team of 4 from 9 players is $C(9, 4) = 126$. Four scoops from 5 flavors with repeats is $C(8, 4) = 70$. Clue words for order: “arrange”, “password”, “first, second, third”; for no order: “choose”, “committee”, “team”.

3.3 Pigeonhole principle

- Need to understand: if n pigeons go into m holes with $n > m$, some hole has at least 2; in general some hole has at least $\lceil n / m \rceil$. To guarantee k in one hole you need $m(k - 1) + 1$ pigeons. First decide what the pigeons and the holes are.
- Example: 12 months, so 37 friends guarantee 4 with a birthday in the same month ($12 \times 3 + 1$).

3.4 Probability

- Need to know that with equally likely outcomes $P(E) = |E| / |S|$; count both with the multiplication principle or combinations. Examples: two dice, sum 4: $3/36 = 1/12$. Five coins, exactly 2 heads: $C(5, 2) / 2^5 = 10/32$.
- **Important:** for “2 red AND 2 green” from 6 red and 4 green balls, multiply: $C(6, 2) \times C(4, 2) / C(10, 4) = 15 \times 6 / 210 = 3/7$.

3.5 Recurrence relations

- Need to understand that a recurrence is an equation plus an initial condition. Example: $a_n = a_{n-1} + 3$ with $a_1 = 4$ gives 4, 7, 10, 13, …; Fibonacci $f_n = f_{n-1} + f_{n-2}$ gives 1, 1, 2, 3, 5, 8, ….
- Need to be able to find an explicit formula by writing out terms and spotting the pattern: $a_n = 4 + 3(n - 1) = 3n + 1$. Compound interest: $A_n = (1.05)^n A_0$.